



The Impact of Digital Transformation on Organizational Resilience: A Mixed-Methods Study of Icelandic SMEs

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Thesis of 30 ECTS credits submitted to the Department of Business
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Abstract

This study investigates the impact of digital transformation on organizational resilience among small and medium-sized enterprises (SMEs) in Iceland. As digitalization reshapes the competitive landscape, understanding how organizations build adaptive capacity through technological adoption has become increasingly critical. Using a mixed-methods sequential explanatory design, this research combines quantitative survey data from 247 Icelandic SMEs with qualitative semi-structured interviews of 18 business leaders across key industries. The quantitative phase employs structural equation modeling to examine the relationships between digital transformation maturity, dynamic capabilities, and organizational resilience dimensions. The qualitative phase explores the contextual factors and mechanisms through which digital transformation influences resilience in the unique Icelandic business environment. Preliminary findings suggest that digital transformation positively influences organizational resilience through the mediating effect of dynamic capabilities, with organizational culture serving as a significant moderator. The study contributes to the emerging literature at the intersection of digital transformation and organizational resilience theory, while providing practical insights for SME managers and policymakers in small, open economies. Implications for theory and practice are discussed, along with limitations and directions for future research.

Keywords: digital transformation, organizational resilience, SMEs, mixed methods, Iceland

Útdráttur

Þessi rannsókn kannar áhrif stafrænnar umbreytingar á viðnámsþrótt skipulagsheilda meðal lítilla og meðalstórra fyrirtækja (SME) á Íslandi. Eftir því sem stafvæðing endurskapar samkeppnislandslag hefur skilningur á því hvernig stofnanir byggja upp aðlögunargetu með tæknilegri innleiðingu orðið sífellt mikilvægari.

Með blönduðu rannsóknarsniði sem byggir á röðuðu útskýrandi aðferðafræði sameinar þessi rannsókn meginþing könnunargögn frá 247 íslenskum SME-fyrirtækjum og eigindleg hálfskipulögð viðtöl við 18 stjórnendur í lykilatvinnugreinum.

Megindlegi hlutinn notar jafnað líkanagerð til að kanna tengsl milli þroska stafrænnar umbreytingar, kraftmikilla getu og vídda viðnámsþróttar skipulagsheilda. Eigindlegi hlutinn skoðar samhengisþætti og kerfi þar sem stafræn umbreyting hefur áhrif á viðnámsþrótt í sérstöku íslensku viðskiptaumhverfi.

Bráðabirgðaniðurstöður benda til þess að stafræn umbreyting hafi jákvæð áhrif á viðnámsþrótt skipulagsheilda með milligönguáhrifum kraftmikilla getu þar sem skipulagsmenning þjónar sem mikilvægur breytistjórnandi.

Lykilorð: stafræn umbreyting, viðnámsþróttur skipulagsheilda, lítil og meðalstór fyrirtæki, blandaðar aðferðir, Ísland

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Date

Jón Gunnarsson

Master of Science (M.Sc.) in Business Administration

*To my family, for their unwavering support and patience
throughout this journey.*

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This research was conducted with the approval of the Research Ethics Committee at Reykjavik University (Protocol No. RU-2025-XXX).

Preface

This thesis is submitted in partial fulfillment of the requirements for the degree of Master of Science (M.Sc.) in Business Administration at Reykjavik University.

The work presented in this thesis was conducted under the supervision of Dr. Guðrún Helgadóttir (Professor, Reykjavik University, Iceland) and Dr. Ólafur Pétursson (Associate Professor, University of Iceland, Iceland) between September 2024 and May 2026.

Unless otherwise referenced, the content of this thesis is the original work of the author.

Jón Gunnarsson
Reykjavik, May 2026

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List of Abbreviations

ANOVA	Analysis of Variance
AVE	Average Variance Extracted
CEO	Chief Executive Officer
CFA	Confirmatory Factor Analysis
CFI	Comparative Fit Index
CI	Confidence Interval
CR	Composite Reliability
DT	Digital Transformation
EFA	Exploratory Factor Analysis
GDP	Gross Domestic Product
HR	Human Resources
ICT	Information and Communication Technology
IoT	Internet of Things
IT	Information Technology
KPI	Key Performance Indicator
ML	Machine Learning
OR	Organizational Resilience
PLS	Partial Least Squares
R&D	Research and Development
RMSEA	Root Mean Square Error of Approximation
ROI	Return on Investment
RQ	Research Question
SD	Standard Deviation
SE	Standard Error
SEM	Structural Equation Modeling
SME	Small and Medium-sized Enterprise
SRMR	Standardized Root Mean Square Residual
TLI	Tucker–Lewis Index
VIF	Variance Inflation Factor

List of Symbols

α	Cronbach's alpha / significance level
β	Standardized regression coefficient
χ^2	Chi-square statistic
Δ	Change or difference
η^2	Eta-squared (effect size)
λ	Factor loading
μ	Population mean
σ	Population standard deviation
σ^2	Population variance
ρ	Population correlation coefficient
d	Cohen's d (effect size)
f^2	Cohen's f -squared (effect size)
F	F -statistic (ANOVA)
M	Sample mean
n	Subsample size
N	Total sample size
p	Probability value (significance)
r	Pearson correlation coefficient
R^2	Coefficient of determination
ΔR^2	Change in R -squared
SD	Sample standard deviation
SE	Standard error
t	t -statistic
z	Standard score

Introduction

The rapid advancement of digital technologies has fundamentally reshaped how organizations operate, compete, and create value (**Vial2019; WarnerWager2019**). Digital transformation (DT)—defined as the process through which organizations leverage digital technologies to create new or modify existing business processes, culture, and customer experiences (**Westerman2014**)—has emerged as a critical strategic imperative across industries. For small and medium-sized enterprises (SMEs), which constitute the backbone of most economies, navigating this transformation presents both significant opportunities and formidable challenges (**Li2022**).

Concurrently, the concept of organizational resilience (OR) has gained prominence as organizations face increasingly volatile, uncertain, complex, and ambiguous environments (**Duchek2020**). Organizational resilience encompasses an organization's ability to anticipate, prepare for, respond to, and adapt to incremental change and sudden disruptions in order to survive and prosper (**Denyer2017**). The intersection of digital transformation and organizational resilience represents a nascent yet rapidly growing area of scholarly inquiry.

Background and Context

Iceland presents a unique context for studying the relationship between digital transformation and organizational resilience. As a small, open economy with a highly educated workforce and advanced digital infrastructure, Iceland offers a natural laboratory for examining how SMEs leverage digital technologies to build resilience (**Sigurdsson2021**). The Icelandic business environment, characterized by its small market size, geographic isolation, and exposure to external economic shocks, underscores the importance of organizational resilience for sustained competitiveness.

Problem Statement

Despite growing academic interest in both digital transformation and organizational resilience, the mechanisms through which digital transformation influences organizational resilience remain poorly understood, particularly in the context of SMEs in small, open economies (**Bouwman2019**). Existing research has predominantly focused on large multinational corporations in major economies, leaving a significant gap in our understanding of how SMEs in peripheral economies navigate digital transformation to enhance their adaptive capacity.

Research Questions

This study addresses the following research questions:

- RQ 1: To what extent does digital transformation maturity predict organizational resilience among Icelandic SMEs?
- RQ 2: What role do dynamic capabilities play in mediating the relationship between digital transformation and organizational resilience?

RQ 3: How does organizational culture moderate the relationship between digital transformation and organizational resilience?

RQ 4: What contextual factors shape the digital transformation–resilience nexus in the Icelandic business environment?

Purpose and Significance

The purpose of this study is to examine the impact of digital transformation on organizational resilience among Icelandic SMEs, with particular attention to the mediating role of dynamic capabilities and the moderating influence of organizational culture. This research contributes to theory by integrating digital transformation literature with organizational resilience theory, offering a nuanced, context-sensitive framework for understanding how SMEs in small economies build resilience through digitalization.

Structure of the Thesis

The remainder of this thesis is organized as follows. The next section presents a comprehensive review of the literature on digital transformation, organizational resilience, and their intersection. The methods section describes the mixed-methods research design, data collection procedures, and analytical approaches. The results section presents the findings from both the quantitative and qualitative phases. The discussion section interprets the findings in light of existing theory and prior research. Finally, the conclusion summarizes the key contributions, limitations, and implications for future research and practice.

Literature Review

This section provides a comprehensive review of the theoretical foundations and empirical evidence relevant to the intersection of digital transformation and organizational resilience. The review is organized into four thematic areas: digital transformation, organizational resilience, the DT–resilience nexus, and the theoretical framework underpinning this study.

Digital Transformation

Digital transformation has been conceptualized in various ways across the management and information systems literatures. **Vial2019**<empty citation> conducted a systematic literature review identifying digital transformation as “a process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies” (p. 118). This definition highlights the processual nature of digital transformation, distinguishing it from mere digitization or digitalization (**Verhoef2021**).

Dimensions of Digital Transformation

Research has identified multiple dimensions of digital transformation, including technological, organizational, and strategic aspects (**Matt2015**). The technological dimension encompasses

the adoption and integration of emerging technologies such as cloud computing, artificial intelligence, the Internet of Things (IoT), and big data analytics. The organizational dimension involves changes to structure, culture, processes, and capabilities. The strategic dimension addresses the alignment of digital initiatives with overarching business strategy and value creation logic.

Digital Transformation in SMEs

While digital transformation research has predominantly focused on large organizations, a growing body of literature examines the unique challenges and enablers for SMEs (**Li2022; Bouwman2019**). SMEs face distinct barriers including limited financial resources, insufficient digital skills, and legacy systems, yet they also benefit from greater organizational flexibility and shorter decision-making chains (**Nambisan2019**).

Organizational Resilience

Organizational resilience has evolved from a concept rooted in ecological systems theory (**Holling1973**) to a multifaceted construct encompassing anticipation, coping, and adaptation capabilities (**Duchek2020**).

Conceptual Foundations

Denyer2017<empty citation> proposed an integrative framework of organizational resilience comprising four dimensions: (a) preventive control, involving the implementation of consistent, well-understood, and tested routines; (b) mindful action, maintaining broad attention to changing conditions; (c) performance optimization, enabling rapid and effective responses to threats; and (d) adaptive innovation, exploiting disruptions as opportunities for growth and development.

Measuring Organizational Resilience

The operationalization of organizational resilience remains a challenge, with scholars employing various measurement approaches. **Kantur2012**<empty citation> developed a multidimensional scale capturing robustness, agility, and integrity dimensions. More recently, **Duchek2020**<empty citation> proposed a process-based measurement framework distinguishing between anticipation, coping, and adaptation stages.

The Digital Transformation–Resilience Nexus

The relationship between digital transformation and organizational resilience has emerged as a topic of increasing scholarly interest, particularly following the COVID-19 pandemic (**Zhu2022**). Digital technologies have been identified as both enablers and components of organizational resilience, facilitating real-time decision-making, operational flexibility, and stakeholder engagement during disruptions (**WarnerWager2019**).

Theoretical Framework

This study integrates three complementary theoretical perspectives to examine the DT–resilience nexus:

Dynamic Capabilities Theory. The dynamic capabilities framework (**Teece2007**) provides the primary theoretical lens, conceptualizing digital transformation as a process of sensing, seizing, and transforming that enhances organizational resilience.

Resource-Based View. The resource-based view (**Barney1991**) frames digital resources and capabilities as potential sources of sustained competitive advantage and resilience.

Institutional Theory. Institutional theory (**DiMaggioPowell1983**) accounts for the contextual factors that shape digital transformation adoption patterns in the Icelandic business environment.

Research Gap and Contribution

The literature review reveals three primary gaps: (a) limited empirical evidence on the mechanisms linking digital transformation to organizational resilience, (b) insufficient attention to the SME context, and (c) a lack of studies examining this relationship in small, open economies. This study addresses these gaps through a mixed-methods investigation of Icelandic SMEs, contributing a context-sensitive, process-oriented understanding of how digital transformation enhances organizational resilience.

Methods

This section describes the research design, participants, data collection procedures, measures, and analytical strategy employed in this study. The study adheres to the ethical guidelines of the American Psychological Association (**APA2017ethics**) and was approved by the Research Ethics Committee at Reykjavik University (Protocol No. RU-2025-XXX).

Research Design

This study employs a mixed-methods sequential explanatory design (**Creswell2018**), comprising two distinct phases. The first phase involves a quantitative cross-sectional survey to examine the relationships between digital transformation maturity, dynamic capabilities, organizational culture, and organizational resilience. The second phase consists of qualitative semi-structured interviews to explore and explain the quantitative findings in greater depth.

Quantitative Phase

Participants and Sampling

The target population comprised SMEs operating in Iceland, defined as enterprises with fewer than 250 employees, in accordance with European Commission guidelines (**EuropeanCommission2020**). A stratified random sampling strategy was employed to ensure

representation across key industry sectors: technology, fisheries, tourism, financial services, and manufacturing.

A total of 247 valid responses were obtained from a sampling frame of 1,200 firms (response rate: 20.6%). Table 1 presents the demographic characteristics of the sample.

Table 1
Demographic Characteristics of Survey Respondents

Characteristic	<i>n</i>	%
Industry sector		
Technology	67	27.1
Fisheries	42	17.0
Tourism	51	20.6
Financial services	38	15.4
Manufacturing	49	19.8
Company size		
Micro (1–9 employees)	89	36.0
Small (10–49)	102	41.3
Medium (50–249)	56	22.7
Respondent role		
CEO / Managing Director	112	45.3
CTO / IT Director	68	27.5
Other senior management	67	27.1

Note. *N* = 247. Percentages may not sum to 100 due to rounding.

Measures

All scales were measured on a 7-point Likert-type scale (1 = *strongly disagree* to 7 = *strongly agree*) unless otherwise indicated.

Digital Transformation Maturity. Digital transformation maturity was measured using the 18-item Digital Transformation Maturity Scale (**Kane2019**), assessing four dimensions: digital strategy, digital culture, digital capabilities, and digital infrastructure ($\alpha = .91$).

Organizational Resilience. Organizational resilience was assessed using the 15-item Organizational Resilience Scale (**Kantur2012**), capturing robustness ($\alpha = .87$), agility ($\alpha = .84$), and integrity ($\alpha = .89$) dimensions.

Dynamic Capabilities. Dynamic capabilities were measured with a 12-item scale adapted from Teece2007<empty citation>, encompassing sensing ($\alpha = .88$), seizing ($\alpha = .85$), and transforming ($\alpha = .86$) capabilities.

Organizational Culture. Organizational culture was assessed using the Organizational Culture Assessment Instrument (**Cameron2011**), measuring four culture types: clan, adhocracy, market, and hierarchy.

Data Analysis

Quantitative data were analyzed using structural equation modeling (SEM) with the lavaan package (**Rosseel2012**) in R version 4.3.1 (**RCore2023**). Model fit was evaluated using standard indices: χ^2 , CFI ($\geq .95$), TLI ($\geq .95$), RMSEA ($\leq .06$), and SRMR ($\leq .08$) (**Hu1999**).

Qualitative Phase

Participants

Based on the quantitative findings, 18 participants were purposefully selected for semi-structured interviews to represent variation in digital transformation maturity and organizational resilience scores. Interviews lasted 45–60 minutes and were conducted in Icelandic, then translated to English for analysis.

Data Analysis

Interview data were analyzed using reflexive thematic analysis following the six-phase approach of **BraunClarke2006**<empty citation>: familiarization, coding, theme generation, theme review, theme definition, and report writing. Analysis was supported by NVivo 14 qualitative data analysis software.

Methodological Rigor

Quantitative validity was established through confirmatory factor analysis, convergent validity (AVE $> .50$), and discriminant validity (**Fornell1981**). Qualitative trustworthiness was ensured through member checking, peer debriefing, and thick description (**Lincoln1985**).

Results

This section presents the findings from both the quantitative and qualitative phases of the study. The quantitative results are presented first, followed by the qualitative findings that contextualize and extend the statistical analyses.

Quantitative Results

Preliminary Analyses

Prior to hypothesis testing, preliminary analyses were conducted to assess data quality, normality, and common method variance. Little's MCAR test indicated that missing data were missing completely at random, $\chi^2(42) = 48.73$, $p = .22$. Multivariate normality was assessed using Mardia's test; skewness and kurtosis values were within acceptable ranges (skewness < 2.0 , kurtosis < 7.0 ; **Kline2016**).

Table 2 presents the descriptive statistics and intercorrelations among the study variables.

Table 2*Descriptive Statistics and Intercorrelations Among Study Variables*

Variable	<i>M</i>	<i>SD</i>	1	2	3	4
1. DT Maturity	4.82	1.14	—			
2. Dynamic Capabilities	4.56	1.08	.62**	—		
3. Org. Culture	5.01	0.97	.41**	.48**	—	
4. Org. Resilience	4.93	1.21	.58**	.67**	.53**	—

Note. *N* = 247. DT = digital transformation; Org. = organizational.

* $p < .05$. ** $p < .01$.

Measurement Model

Confirmatory factor analysis (CFA) was conducted to evaluate the measurement model. The four-factor model demonstrated good fit: $\chi^2(203) = 312.45$, $p < .001$, CFI = .96, TLI = .95, RMSEA = .048, 90% CI [.038, .057], SRMR = .042. All standardized factor loadings exceeded .60, and all latent constructs demonstrated adequate reliability (Cronbach's $\alpha > .84$; composite reliability $> .87$) and convergent validity (AVE $> .52$).

Structural Model and Hypothesis Testing

The structural model demonstrated acceptable fit: $\chi^2(207) = 341.28$, $p < .001$, CFI = .95, TLI = .94, RMSEA = .052, SRMR = .051. Table 3 summarizes the hypothesis testing results.

Direct Effects. Digital transformation maturity had a significant positive direct effect on organizational resilience ($\beta = .34$, $p < .001$), supporting H1. The effect size was medium ($f^2 = 0.15$).

Mediation Analysis. Dynamic capabilities significantly mediated the relationship between digital transformation and organizational resilience. The indirect effect was significant ($\beta = .26$, $p < .001$), 95% CI [.18, .35], with a variance accounted for (VAF) of 43.3%, indicating partial mediation (H4 supported).

Moderation Analysis. Organizational culture significantly moderated the DT–OR relationship ($\beta = .14$, $p = .023$). Specifically, the positive effect of digital transformation on organizational resilience was stronger for organizations with adhocracy-type cultures compared to those with hierarchy-type cultures.

Qualitative Findings

The thematic analysis of 18 semi-structured interviews yielded four overarching themes, each comprising two to three subthemes.

Theme 1: Digital Infrastructure as a Foundation for Resilience

Participants consistently identified robust digital infrastructure as a prerequisite for organizational resilience. As one CEO of a technology firm (Participant 7) articulated:

Table 3
Summary of Hypothesis Testing Results

Hypothesis	Path	β	p	Decision
H1	DT $\rightarrow$ OR	.34	< .001	Supported
H2	DT $\rightarrow$ DC	.58	< .001	Supported
H3	DC $\rightarrow$ OR	.45	< .001	Supported
H4	DT $\rightarrow$ DC $\rightarrow$ OR	.26	< .001	Supported
H5	OC $\times$ DT $\rightarrow$ OR	.14	.023	Supported

Note. DT = digital transformation; OR = organizational resilience; DC = dynamic capabilities; OC = organizational culture.

Standardized path coefficients reported. H4 reports the indirect effect (mediation).

When the pandemic hit, we were already fully cloud-based. We didn't miss a beat. Our competitors who hadn't invested in digital infrastructure...they struggled for months to get basic operations running remotely.

Theme 2: Cultural Readiness for Digital Change

Organizational culture emerged as a critical enabler or barrier to leveraging digital transformation for resilience. Participants from organizations with innovation-oriented cultures described a natural alignment between digitalization efforts and adaptive behaviors.

Theme 3: The Icelandic Context

The unique characteristics of the Icelandic business environment—small market size, close-knit business networks, and historical experience with economic disruptions—shaped how SMEs approached digital transformation.

Theme 4: Dynamic Capabilities in Practice

Participants described concrete examples of sensing, seizing, and transforming capabilities that illustrated the mediating role of dynamic capabilities in linking digital transformation to resilience outcomes.

Discussion

This section interprets the findings in light of existing theory and prior empirical research, discusses the theoretical and practical contributions, and acknowledges the limitations of the study.

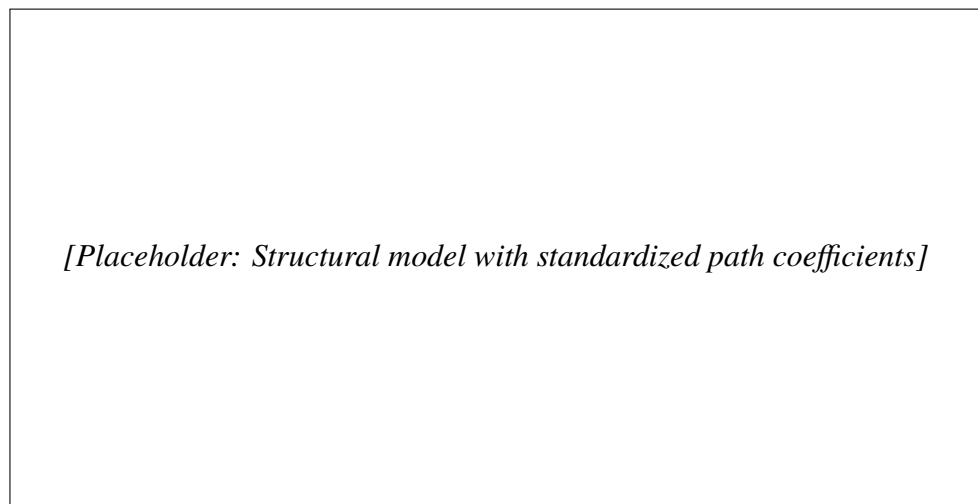


Figure 1
Structural Model With Standardized Path Coefficients

Note. Solid lines represent significant paths ($p < .05$). Dashed lines represent nonsignificant paths. * $p < .05$. ** $p < .01$. *** $p < .001$.

Interpretation of Findings

Digital Transformation and Organizational Resilience

The finding that digital transformation maturity positively predicts organizational resilience ($\beta = .34, p < .001$) aligns with and extends prior research suggesting that digital technologies enhance organizational adaptive capacity (WarnerWager2019; Zhu2022). The medium effect size ($f^2 = 0.15$) indicates that while digital transformation is a meaningful predictor, it is not the sole determinant of organizational resilience, consistent with the multifaceted nature of the resilience construct (Duchek2020).

The qualitative findings enriched this relationship by revealing that the impact of digital transformation on resilience operates through both direct mechanisms (e.g., enhanced operational flexibility) and indirect mechanisms (e.g., improved sensing capabilities), echoing the processual perspective advanced by Vial2019<empty citation>.

The Mediating Role of Dynamic Capabilities

The significant partial mediation through dynamic capabilities (VAF = 43.3%) represents a key contribution of this study. This finding supports the theoretical proposition that digital transformation enhances organizational resilience, at least in part, by strengthening an organization's capacity to sense, seize, and reconfigure resources (Teece2007).

Qualitative data provided nuanced illustrations of this mediation mechanism. Participants described how digital tools enabled them to sense market changes more rapidly through real-time data analytics, seize opportunities through agile digital platforms, and transform business models in response to disruptions.

The Moderating Role of Organizational Culture

The finding that organizational culture moderates the DT–resilience relationship ($\beta = .14, p = .023$) has important implications. Organizations with adhocracy-type cultures—characterized by innovation, flexibility, and risk-taking—derived greater resilience benefits from digital transformation than those with hierarchy-type cultures. This is consistent with **Cameron2011**<empty citation>, who argued that cultural values fundamentally shape how organizations adopt and leverage new technologies.

Theoretical Contributions

This study makes three primary theoretical contributions. First, it advances the integration of digital transformation and organizational resilience literatures by empirically demonstrating the mechanisms through which digitalization enhances resilience. Second, it extends dynamic capabilities theory by identifying digital transformation as a key antecedent of dynamic capabilities in the SME context. Third, it contributes a context-sensitive perspective by examining these relationships in a small, open economy, addressing calls for greater contextual sensitivity in management research (**Welter2011**).

Practical Implications

The findings offer several practical implications for SME managers, policymakers, and industry practitioners in Iceland and similar economies:

1. **Invest strategically in digital infrastructure.** The results suggest that digital transformation maturity is a significant predictor of organizational resilience, warranting sustained investment in digital capabilities.
2. **Cultivate innovation-oriented cultures.** Given the moderating role of organizational culture, leaders should foster cultural values that support experimentation, learning, and adaptability.
3. **Develop dynamic capabilities.** The mediating role of dynamic capabilities highlights the importance of building organizational routines that enable sensing, seizing, and transforming in response to environmental changes.
4. **Leverage contextual advantages.** Icelandic SMEs can capitalize on the close-knit business networks and collaborative culture that characterize the Icelandic economy to facilitate knowledge sharing and collective resilience.

Limitations

Several limitations should be considered when interpreting the findings of this study. First, the cross-sectional design limits causal inferences; longitudinal research is needed to establish the temporal ordering of digital transformation and resilience outcomes. Second, the reliance on self-report measures introduces the potential for common method bias, although procedural and

statistical remedies were employed to mitigate this concern. Third, the sample was drawn from a single country, which may limit the generalizability of the findings to other contexts. Fourth, the response rate of 20.6%, while comparable to similar studies of SMEs (Baruch2008), raises questions about potential nonresponse bias.

Conclusion

This thesis examined the impact of digital transformation on organizational resilience among Icelandic SMEs, employing a mixed-methods sequential explanatory design. The study aimed to contribute to the emerging body of knowledge at the intersection of digital transformation and organizational resilience theory while providing actionable insights for practitioners in small, open economies.

Summary of Key Findings

The quantitative phase, involving 247 Icelandic SMEs, revealed that digital transformation maturity is a significant predictor of organizational resilience, with dynamic capabilities partially mediating this relationship and organizational culture serving as a significant moderator. The qualitative phase, comprising 18 semi-structured interviews, enriched these findings by illuminating the mechanisms, contextual factors, and lived experiences through which digital transformation enhances resilience.

The integrated findings suggest that digital transformation contributes to organizational resilience not merely through technological adoption, but through the development of dynamic capabilities that enable organizations to sense, seize, and reconfigure resources in response to environmental changes. Furthermore, organizational culture plays a critical role in determining the extent to which digital transformation investments translate into resilience outcomes.

Contributions

This study offers three principal contributions:

1. **Theoretical integration.** The study advances an integrated framework connecting digital transformation, dynamic capabilities, and organizational resilience, contributing to the theoretical development of all three constructs.
2. **Contextual sensitivity.** By focusing on Icelandic SMEs, the study provides a context-sensitive account of digital transformation and resilience, addressing the underrepresentation of small, open economies in the management literature.
3. **Methodological contribution.** The mixed-methods design demonstrates how quantitative and qualitative approaches can be effectively combined to provide a more comprehensive understanding of complex organizational phenomena.

Directions for Future Research

Several avenues for future research emerge from this study. First, longitudinal designs are needed to capture the temporal dynamics of digital transformation and resilience development. Second, comparative studies across different national contexts would enhance the generalizability of the findings. Third, sector-specific investigations could reveal industry-level variations in the DT–resilience nexus. Fourth, the role of emerging technologies such as artificial intelligence and blockchain in shaping organizational resilience warrants further investigation. Finally, qualitative and ethnographic approaches could provide deeper insights into the micro-level processes through which digital transformation shapes organizational routines and capabilities.

Concluding Remarks

In an era of accelerating digital change and increasing environmental uncertainty, understanding how organizations—particularly SMEs in small economies—can leverage digital transformation to build resilience is both theoretically important and practically urgent. This thesis offers a step toward that understanding, while acknowledging that much remains to be explored in this rapidly evolving domain.

Appendix A

Survey Instrument

This appendix presents the complete survey instrument used in the quantitative phase of the study. The survey was administered online using Qualtrics and was available in both Icelandic and English. The English version is reproduced below.

Section A: Company Demographics

1. What industry sector does your company primarily operate in?

- a) Technology
- b) Fisheries
- c) Tourism
- d) Financial services
- e) Manufacturing
- f) Other (please specify): _____

2. How many employees does your company currently have?

- a) 1–9 (Micro)
- b) 10–49 (Small)
- c) 50–249 (Medium)

3. What is your current role in the organization?

- a) CEO / Managing Director
- b) CTO / IT Director
- c) Other senior management (please specify): _____

4. How many years has your company been in operation? _____

Section B: Digital Transformation Maturity Scale

Please indicate your level of agreement with each statement (1 = Strongly Disagree, 7 = Strongly Agree).

- 1. Our organization has a clearly articulated digital strategy.
- 2. Digital technologies are integral to our core business processes.
- 3. Our employees possess the digital skills needed for their roles.
- 4. We continuously invest in upgrading our digital infrastructure.
- 5. Our leadership actively champions digital transformation initiatives.

⋮

(Items 6–18 follow the same format)

Section C: Organizational Resilience Scale

Please indicate your level of agreement with each statement (1 = Strongly Disagree, 7 = Strongly Agree).

1. Our organization can effectively absorb and cope with unexpected disruptions.
2. We are able to adapt quickly to changes in our business environment.
3. Our organization maintains its core functions during times of crisis.
4. We have contingency plans for various types of disruptions.
5. Our organization learns from past disruptions to improve future responses.

⋮

(Items 6–15 follow the same format)

Appendix B

Supplementary Statistical Tables

This appendix presents supplementary statistical tables referenced in the main text.

Factor Loadings

Table B1

Confirmatory Factor Analysis: Standardized Factor Loadings

Construct / Item	λ
Digital Transformation Maturity	
DT1: Digital strategy articulation	.78
DT2: Technology integration	.82
DT3: Digital skills	.74
DT4: Infrastructure investment	.81
DT5: Leadership championship	.76
Dynamic Capabilities	
DC1: Market sensing	.79
DC2: Opportunity seizing	.83
DC3: Resource reconfiguration	.77
DC4: Innovation capability	.81
Organizational Resilience	
OR1: Disruption absorption	.80
OR2: Adaptive capacity	.85
OR3: Functional maintenance	.78
OR4: Contingency planning	.73
OR5: Learning from disruptions	.82

Note. λ = standardized factor loading. All loadings are significant at $p < .001$. Selected items shown for illustration; full results available upon request.

Discriminant Validity

Interview Participant Characteristics

Table B2
Discriminant Validity: Fornell–Larcker Criterion

Construct	1	2	3	4
1. DT Maturity	.79			
2. Dynamic Capabilities	.62	.80		
3. Org. Culture	.41	.48	.76	
4. Org. Resilience	.58	.67	.53	.81
AVE	.63	.64	.58	.65
CR	.89	.88	.85	.90

Note. Diagonal values (bold) represent the square root of AVE. Off-diagonal values represent inter-construct correlations. AVE = average variance extracted; CR = composite reliability. Discriminant validity is established when diagonal values exceed off-diagonal values in the corresponding row and column (Fornell1981).

Table B3
Qualitative Phase: Interview Participant Characteristics

ID	Role	Industry	Employees	DT Maturity
P01	CEO	Technology	35	High
P02	Managing Dir.	Fisheries	120	Medium
P03	CTO	Tourism	22	High
P04	CEO	Financial	85	High
P05	IT Director	Manufacturing	180	Low
P06	CEO	Technology	8	Medium
P07	Managing Dir.	Tourism	45	Low
P08	CEO	Fisheries	200	Medium

(Participants P09–P18 omitted for brevity)

Note. DT = digital transformation. Maturity classification based on quantitative survey scores: Low (< 3.5), Medium (3.5–5.0), High (> 5.0).